

pharmsol: A high-performance Rust library for pharmacokinetic and pharmacodynamic modeling and simulation

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Summary

pharmsol is a library for pharmacokinetic/pharmacodynamic (PK/PD) simulation written in Rust. It provides the necessary tools and frameworks for defining, solving, and analyzing compartmental models, with support for differential equations, their analytical solutions, and experimental support for stochastic differential equations. Written in Rust, the library aims to provide researchers and developers with a framework for pharmacokinetic simulation in a memory-safe and performant language. The library is distributed via crates.io with comprehensive API documentation, usage examples, and a test suite validated through continuous integration.

Statement of need

Pharmacokinetic and pharmacodynamic modeling and simulation are computationally intense when applied to modern, complex, and sophisticated dosing regimens, mechanistic models, and individualized approaches. Unlike comprehensive pharmacometric platforms such as NONMEM (Beal & Sheiner, 1992), Phoenix NLME (Certara USA, Inc., 2024), or Monolix (Simulations Plus (United States), 2024), pharmsol is purpose-built as a simulation engine that pharmacometricians can leverage to rapidly execute simulations for individuals or populations with pre- and user-defined models.

As a fully open-source solution, pharmsol empowers users to inspect, modify, and extend the simulation capabilities without licensing constraints. Users can define custom models by specifying their own differential equations as closures, or use the provided analytical solutions for standard compartmental models. Additionally, pharmsol can be integrated in more user-friendly languages such as R using extendr (Reimert et al., 2024), making it accessible to pharmacometricians who may prefer higher-level interfaces.

State of the field

Several tools exist for performing pharmacokinetic and pharmacodynamic modelling and simulation. Most well known are the full-suite applications such as NONMEM (Beal & Sheiner, 1992), Phoenix NLME (Certara USA, Inc., 2024), Monolix (Simulations Plus (United States), 2024), and Pumas (Rackauckas et al., 2020). These are generally commercial or proprietary, while open-source alternatives include the R packages Pmetrics (Neely et al., 2012) and mrgsolve (Baron, 2025), as well as implementations in Python and Julia. The latter two are powered by Fortran and C++, respectively.

39 The choice to develop `pharmsol` as a new library rather than contribute to the existing solutions
40 is motivated by several factors. First, Rust provides compile-time memory safety guarantees
41 which are not available in Fortran or C++, without sacrificing performance. This is especially
42 important for software used in healthcare settings. Second, existing alternatives are coupled to
43 their parent frameworks, and may be difficult to integrate with new solutions or software.

44 The aim of `pharmsol` is to provide developers of higher-level software with a library for
45 performing efficient and memory-safe pharmacokinetic modeling and simulation. Importantly,
46 `pharmsol` is open-source with a permissive license, allowing developers to inspect, extend, and
47 integrate the library into new or existing solutions.

48 Software design

49 To the authors' knowledge, `pharmsol` is the first software written in Rust for pharmacokinetic
50 modeling and simulation. The design principle of `pharmsol` is to provide researchers and
51 pharmacometricians a flexible and efficient library for solving pharmacokinetic problems.

52 These problems are defined by two key aspects: the pharmacokinetic data, and the structural
53 model. While the data structure is explained in detail below, the structural model is more
54 complex. In `pharmsol`, models are defined using a set of closures, providing methods to define
55 the set of equations that describe the mass transfer of drug, as well as initial conditions and
56 output equations, i.e. detailing the initial and observed states of the system. Bioavailability
57 terms and lag-time of absorption are also supported. Covariates may be applied on both
58 parameters and state values. Importantly, `pharmsol` provides both linear interpolation and
59 carry-forward for covariates, both common in population pharmacokinetic models.

60 The three different solver types, i.e. analytical, ODE, and SDE based, are all built around a
61 common Equation trait. This gives `pharmsol` a modular design, allowing users to easily extend
62 the library with new solvers or to rely on it while being agnostic on the underlying solver type.
63 This flexibility is particularly useful when integrating `pharmsol` with other software, as it allows
64 users to select the most appropriate solver for their specific use case (e.g., speed vs. model
65 complexity).

66 Research impact statement

67 As a tool for pharmacokinetic modelling and simulation, `pharmsol` has already demonstrated
68 research impact. Most importantly, it is used as the simulation engine for `PMcore`, the Rust
69 implementation of `Pmetrics` (Neely et al., 2012) - the *de facto* R package for non-parametric
70 population pharmacokinetic modelling and simulation. In this regard, `pharmsol` has been
71 used to generate results in scientific publications, such as the creation of a new algorithm for
72 non-parametric parameter estimation (Hovd et al., 2025).

73 By utilizing the SDE solver in `pharmsol`, our research group is currently researching the use
74 of SDEs in a non-parametric pharmacokinetic framework. We have also developed internal
75 applications which rely on `pharmsol` for simulating alternative dose regimens to individualize
76 drug therapy.

77 Data format

78 `pharmsol` is designed around a hierarchical data structure that models the typical organization
79 of pharmacometric data. The primary data struct, `Data`, is a collection of `Subjects`, which
80 may have one or more `Occasions`, i.e. separate pharmacokinetic investigations. Each occasion
81 consists of one or more `Events`, e.g. an instantaneous dose (bolus), infusions of drug, or
82 observed concentrations at given times.

Data → Subject → Occasion → Event (Bolus, Infusion, Observation)

83 Currently, pharmsol provides methods to parse the Pmetrics (Neely et al., 2012) data format.
84 In the future, we aim to also support additional formats, such as those used by NONMEM,
85 Monolix (Simulations Plus (United States), 2024), and more. Data may also be built using a
86 provided builder-pattern, providing a flexible method for defining subjects programmatically.
87 This is especially useful when integrating pharmsol with third-party software, as it allows for
88 reading and generating data that is not in an already supported format.

89 Supported equation formats

90 As stated before, the Equation trait provides the architectural backbone for simulating PK/PD
91 profiles. Pharmsol currently also provides three solvers implementing this trait: analytical
92 solutions, ordinary differential equations, and experimental support for stochastic differential
93 equations.

94 Analytical Solutions

95 For standard compartmental models, pharmsol provides closed-form solutions for one-, two-
96 and three-compartment models, with and without oral absorption. These have been verified
97 against their differential equation counterparts. Benchmarks demonstrate 20-33× speedups
98 compared to equivalent ODE formulations without loss of precision (see repository benchmarks
99 for details). Additional analytical solutions will be added in future versions.

100 Ordinary Differential Equations

101 For more complex or non-standard models, pharmsol supports user-defined ordinary differential
102 equations (ODEs). The numerical integration is performed using the diffsol library (Robinson
103 & Allmont, 2026), which provides efficient BDF solvers suitable for the stiff systems often
104 encountered in pharmacometric modeling.

105 Stochastic Differential Equations

106 Experimental support for stochastic differential equations (SDEs) is available using the Euler-
107 Maruyama method. SDEs allow modeling of within-subject variability as a continuous stochastic
108 process. However, particular care should be taken if applying SDEs in a non-parametric approach
109 to population pharmacokinetic modeling, such as when using the non-parametric adaptive grid
110 algorithm (NPAG) (Yamada et al., 2020) for parameter estimation.

111 Conclusion and Future Work

112 pharmsol aims to support the evolving needs of pharmacometric research by providing a modern,
113 efficient platform that can adapt to the increasing complexity of pharmaceutical development
114 while remaining accessible through its open-source licensing model. Future development will
115 focus on additional analytical model implementations, support for common data formats used
116 by other pharmacometric software, non-compartmental analysis and continued performance
117 improvements.

118 AI usage disclosure

119 The authors declare that generative AI has been used during software development and
120 manuscript preparation. For software development, generative AI has been used to generate
121 code by instruction, refactor existing code, produce tests and write documentation. For the
122 manuscript, generative AI was used to improve the language of the submission. All use

of generative AI was reviewed, edited, and verified by the authors. The authors take full responsibility for the contents of the submission, and for the software within the scopes of the software license.

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